

Insider Trade Disclosure, Market Efficiency, and Liquidity*

ANDREA M. BUFFA[†]

March 2008

Abstract

Is a more transparent market also more efficient and liquid? We address this question by analyzing the impact of mandatory ex-post disclosure of insider trades (as in Section 16(a) of the U.S. Securities Exchange Act) in a dynamic model of strategic risk averse insider trading. We show that trade disclosure reduces informational efficiency of prices and may cause the market to be less liquid. In our model, the informed trader uses a less aggressive trading strategy in a more transparent market (i.e. with trade disclosure) in order to prevent the market maker from inferring perfectly the private information from public records, and to maintain her informational advantage over time. Our result then questions the effectiveness of such securities regulation.

JEL classification: D82, G14, G18.

Keywords: insider trading, market transparency, trade disclosure, market efficiency, liquidity.

*I thank Viral Acharya, Francesca Cornelli, Antonio Mele, Giovanna Nicodano, Guillaume Plantin, Rohit Rahi, Francesco Sangiorgi, and seminar participants at LBS for their comments. In particular, I am very grateful to Francisco Gomes and Raman Uppal for their advice. Any errors or omissions are the responsibility of the author.

[†]London Business School. Mailing address: 6 Sussex Place Regent's Park, London NW1 4SA, United Kingdom. Email: abuffa.phd2007@london.edu.

1 Introduction

Market transparency has been for many years at the heart of policy debates concerning the design of securities markets. However, there is no agreement among financial market regulators on what should be the optimal degree of transparency. In the United States, the Securities and Exchange Commission’s (SEC) view is straightforward: market transparency “plays a fundamental role in the fairness and efficiency of the secondary market, [...] improves the price discovery, fairness, competitiveness and attractiveness of U.S. market” (SEC, 1995). Similarly, the International Organization of Securities Commissions (IOSCO) states that “market transparency – in essence, the widespread availability of information relating to current opportunities to trade and recently completed trades – is generally regarded as central to both the fairness and efficiency of a market, and in particular to its liquidity and quality of price-formation” (IOSCO, 2001). In contrast, in the United Kingdom, the Securities and Investment Board (SIB) has argued that there is “a tradeoff between liquidity and trade transparency” and consequently “transparency should be restricted if this is necessary to assure adequate liquidity” (SIB, 1994).

Differently from previous works on this issue, this paper analyzes ex-post disclosure of corporate insider trades, as a less explored but relevant aspect of market transparency. According to Section 16(a) of the Securities Exchange Act of 1934, insider traders must report their trades to the SEC not later than two business days following the transaction.¹ The rationale for this securities regulation, as recognized by the authority, is to make private information available to all market participants more rapidly, thus increasing price efficiency and market liquidity:

Section 16(a) is likely to provide significant benefits by making information concerning insiders’ transactions in issuer equity securities publicly available sub-

¹Before the introduction of the Sarbanes-Oxley Act (August 2002), this rule required officers and directors to report transactions in equity securities directly with the related issuer within ten days following the end of the month in which the trade had occurred.

stantially sooner than it was before. Making this information available to all investors on a more timely basis should increase market transparency, which will likely enhance market efficiency and liquidity.

— U.S. Securities and Exchange Commission, File No. S7-31-02.

A similar rule is enforced in the U.K., where corporate insiders must inform their company as soon as possible and no later than the fifth business day after a transaction for their own account or on behalf of their spouses and children. In turn, a company must inform the London Stock Exchange (LSE) without delay and no later than the end of the business day following receipt of the information.² Therefore, an improvement in market transparency through the disclosure of insider trades should translate into a higher liquidity and efficiency of the market. In this paper we present a theoretical model to study whether and to which extent this is true.

The natural choice to analyze dynamic strategic insider trading behavior is the rational expectation trading model pioneered by Kyle (1985). We present here an extended version of it in which we introduce risk aversion and mandatory ex-post trade disclosure. Analyzing both a discrete- and a continuous-time framework, we model the regulation by requiring the informed agent to disclose in each trading period the trade she has made in the previous one. Then, the comparison between equilibrium outcomes for the same economy with and without regulation (hereafter referred as transparent and opaque market respectively) allows us to determine its impact on market components.

Our main result is that transparency reduces informational efficiency of prices and may cause the market to be less liquid. The key to understanding this surprising result is to analyze of the following two scenarios. In an opaque market (i.e. without disclosure regulation), as showed by Holden and Subrahmanyam (1994) and Baruch (2002), the risk averse informed

²This implies that information about an insider transaction can reach the market as late as 6 days after the transaction. However, in practice, this information is disclosed faster: as documented by Fidrmuc et al. (2006) the announcement day for most of the directors' dealings (85 % in their sample) coincides with the transaction day or is the following day.

trader chooses to exploit her informational advantage very rapidly to protect herself against future price risk imposed by liquidity traders. This aggressive trading behavior affects positively both market efficiency and liquidity. Indeed, almost all the private information is revealed to the market at the very early stages of the trading rounds, thus strongly reducing the adverse selection problem faced by the market maker. This implies a very liquid market. In a transparent market (i.e. with disclosure regulation), instead, we find an equilibrium in which the aggressiveness of the informed trader is severely reduced. The introduction of the disclosure requirement creates a tradeoff in the informed agent's strategy between future price risk and the revelation of private information through disclosure. In equilibrium the concern for the latter prevails. Consistently with Huddart et al. (2001) the insider plays a mixed strategy by adding a random component to her order flow. This prevents the market maker from inferring perfectly the private information from public records, and allows the insider to maintain her informational advantage over time. A lower market efficiency is caused by the fact that private information is now slowly incorporated into prices. An interesting interpretation of this result is that trade disclosure can be seen as the institutional means that forces the risk averse insider to behave as if she were risk neutral. As a matter of fact, in equilibrium market efficiency evolves over time exactly in the same way it does in a opaque market with risk neutral agents (Kyle, 1985).

More in details, our results can be summarized as follows: when ex-post insider trade disclosure is imposed (i) the insider maintains an informational advantage over the entire trading period despite the disclosure; (ii) market is less efficient at any trading round; (iii) the differential in market efficiency, in terms of how much of the private information is incorporated into prices, even widens if risk aversion and/or the number of trading periods increase; (iv) trading price has a constant volatility over time and information is incorporated into prices at a constant rate; (v) market liquidity is constant over time and (vi) may be lower at the final trading rounds; (vii) if trading is continuous, even when traders are risk neutral market efficiency does not improve. These results thus question the effectiveness of

such securities regulation.

This paper relates to the literature on market transparency, in which the key issue is the tradeoff between (pre- and post-trade) transparency and liquidity. Analytical results (Madhavan, 1995; Pagano and Röell, 1996; Naik et al., 1999; Frutos and Manzano, 2002), laboratory experiments (Bloomfield and O'Hara, 1999; Flood et al., 1999) and natural experiments (Madhavan et al., 2005) have shown that in a dealer market transparency improves informational efficiency but causes opening spreads to widen.³

This paper also relates to the literature on strategic insider trading with mandated disclosure. Fishman and Hagerty (1995) consider a two period model in which an insider may become informed with a certain probability. When the insider becomes informed, she never manipulates the market, while she might do it (imitating an informed insider with good news) if uninformed. John and Narayanan (1997) extend their model by introducing asymmetry in the probability of receiving a good or a bad signal. In this setting also an informed insider may manipulate the market. In both models manipulation is driven by an uninformed insider's attempt to pool with an informed insider and never occurs in equilibrium with good and bad news simultaneously. The most related article to this paper is Huddart et al. (2001). They extend the discrete time model by Kyle (1985) with risk neutral insider trading by adding the trade disclosure constraint. They find that an equilibrium in which the insider adds a noise component to her trading strategy (*dissimulation strategy*) exists and that the regulation accelerates price discovery and increases market depth. Zhang (2004) extends the model by Huddart et al. (2001) by introducing risk aversion and shows that an equilibrium in mixed strategy still exists. Cao and Ma (1999), instead, extend Huddart et al. (2001) by introducing imperfect competition among insiders. In their model market efficiency is unambiguously higher with disclosure, while market liquidity may be lower when insiders' signal are negatively correlated.⁴

³Flood et al. (1999), instead, find that quote transparency reduces both opening spreads and market efficiency.

⁴Cao and Ma (1999) adopt the same signals structure as in Foster and Viswanathan (1996) and use as a

We contribute to the existing literature on market transparency and strategic insider trading by (i) examining a different aspect of transparency (the requirement for corporate insiders to ex-post disclose their trades) which is of primary importance for the design of securities markets; (ii) obtaining close-form solution for the equilibrium of the limiting model of continuous trading; (iii) formally showing the detrimental effect that transparent markets may have on market efficiency and liquidity.

The remainder of the paper is organized as follow. To give an intuition of the main features of the model, in Section 2 we introduce it in a discrete-time framework; then in Section 3 we present the more elegant setup and equilibrium in a continuous-time framework. In Section 4 we discuss the properties of the equilibria. Section 5 outlines the policy implications of our findings and concludes. All the proofs are relegated to the Appendix.

2 A Sequential Auction Equilibrium

Our model is based on Kyle (1985). Consider a single risky asset traded by an informed trader and a number of liquidity traders in N sequential auctions. The time at which the n -th takes place is denoted by t_n . We assume that the sequence of auction dates $\langle t_n \rangle$ partitions the time interval $[0, 1]$. In each period a market maker observes the total order flow and fixes the price at which all orders are filled. All agents are risk averse but the market maker who is risk neutral.⁵ Since only one agent has private information, no uninformed agent enters the market because unwilling to be trading counterpart of someone with a “superior” information endowment.

Let us denote with v the *ex-post* liquidation value of the risky asset, which is assumed to be normally distributed, $v \sim \mathcal{N}(p_0, \Sigma_0)$. The informed trader has perfect information in

benchmark to their continuous time extension the model without trade disclosure by Back et al. (2000).

⁵As argued by Holden and Subrahmanyam (1994), market making is typically performed by large financial institutions, which have large capacity to bear risk. Therefore, their behavior can be modeled by assuming risk neutrality.

the sense she observes at t_0 the liquidation value v . Let Δx_n be the order placed by the monopolistic insider trader at the n -th auction. The aggregate order at the n -th auction by liquidity traders, who trade for liquidity reasons, is denoted by Δu_n . We assume $\Delta u_n \sim \mathcal{N}(0, \sigma_u^2 \Delta t_n) \quad \forall n$, serially uncorrelated and independent of v . Furthermore, let Δy_n be the total order flow observed by the market maker in period n : $\Delta y_n = \Delta x_n + \Delta u_n$. As in Holden and Subrahmanyam (1994) we denote with W_0 the initial wealth of the informed trader and with W_n the wealth at auction n . Moreover, informed trader has negative exponential utility, with risk aversion coefficient A , for terminal wealth (denoted by W_{N+1}):

$$U(W_{N+1}) = -\exp\{-AW_{N+1}\} \quad (1)$$

At the n -th auction the market maker observes only the total order flow and then sets the price p_n by a competitive process, such that the price equals the expected value of v conditional on the information available at that auction.⁶ Mandatory trade disclosure implies that the market maker, after having set the price p_n , but before setting the market price of the subsequent auction, observes the order placed by the informed trader at the n -th auction, Δx_n . Thus, he adjusts his belief of the asset value from p_n to p_n^* , which is defined by the market maker's estimate of the value of the risky asset given all information available after disclosure. We call p_n^* *updating* price, which is not a market price but rather the price that the market maker would have set for the n -th auction if he had known the insider's order at the beginning of the period.

The information of the market maker is represented by a *prior* (pre-disclosure) and an *updating* (post-disclosure) filtrations, denoted respectively by $\mathbb{F}$ and $\mathbb{F}'$, whereas the information of the informed agent is represented by a single filtration $\mathbb{G}$:

$$\begin{aligned} \mathbb{F} &= \{\mathcal{F}_n\}_{n \geq 0} \quad \text{where} \quad \mathcal{F}_n = \sigma(\Delta y_m : 0 \leq m \leq n) \vee \sigma(\Delta x_m : 0 \leq m < n) \\ \mathbb{F}' &= \{\mathcal{F}'_n\}_{n \geq 0} \quad \text{where} \quad \mathcal{F}'_n = \sigma(\Delta y_m, \Delta x_m : 0 \leq m \leq n) \\ \mathbb{G} &= \{\mathcal{G}_n\}_{n \geq 0} \quad \text{where} \quad \mathcal{G}_n = \sigma(v) \vee \sigma(p_m, p_m^* : 0 \leq m < n) \end{aligned}$$

⁶This is equivalent to assuming Bertrand competition among several market makers, which implies that the profits of the market maker(s) are driven to zero.

The monopolistic insider behaves strategically by choosing a $\mathcal{G}$ -mesurable function as optimal trading strategy that maximizes her expected utility conditional on all public and private information. Let $J(W_n)$ denote the indirect utility from W_n . Equilibrium is therefore defined by a market efficiency and a utility-maximization condition, as formalized in the next Section.

2.1 Equilibrium

In this Section we first define the equilibrium notion we are going to adopt (following closely the definition given by Kyle, 1985) and then we provide the equilibrium conditions of the trading game.

Let $\Delta x_n = X_n(\mathcal{G}_n)$ represent the optimal strategy of the insider trader at the n -th auction, and $p_n = P_n(\mathcal{F}_n)$, $p_n^* = P_n^*(\mathcal{F}'_n)$ the optimal strategies of the market maker before and after disclosure at that auction respectively. Then, let us define $X = \langle X_1, \dots, X_N \rangle$, $P = \langle P_1, \dots, P_N \rangle$ and $P^* = \langle P_1^*, \dots, P_N^* \rangle$ as the vectors of strategy functions.

DEFINITION 1: *An equilibrium of the discrete-time trading game is defined as a triple (X, P, P^*) such that the following conditions hold:*

1. UTILITY MAXIMIZATION. *For any $n \in \{1, \dots, N\}$ and for any alternative trading strategy $X' = \langle X'_1, \dots, X'_N \rangle$:*

$$\mathbb{E} \left[J \left(W_{n+1}(X, P, P^*) \right) \middle| \mathcal{G}_n \right] \geq \mathbb{E} \left[J \left(W_{n+1}(X', P, P^*) \right) \middle| \mathcal{G}_n \right] \quad (2)$$

2. MARKET EFFICIENCY. *For any $n \in \{1, \dots, N\}$ the two prices are competitive:*

$$p_n = \mathbb{E}[v | \mathcal{F}_n] \quad (3)$$

$$p_n^* = \mathbb{E}[v | \mathcal{F}'_n] \quad (4)$$

Given the definition of the equilibrium we now restrict our attention on the class of linear equilibria. In particular, we search for an equilibrium in which both the insider trader and the market maker's strategies are linear functions.

As argued by Huddart et al. (2001), trading and pricing strategies characterizing a linear equilibrium in an opaque market (Kyle, 1985; Holden and Subrahmanyam, 1994) can not be part of an equilibrium in a transparent market. To see this, suppose the contrary holds. Suppose the market maker conjectures that insider's demand is just a linear function of the liquidation value, v , of the risky asset. Then, after disclosure of Δx_1 at the end of the first auction, the market maker would be able to invert the liner function, thus perfectly inferring the liquidation value. So, no private information would be left to the insider. As a consequence, at the second auction the market maker would set the price equal to v (choosing zero sensitivity to the order flow in his pricing function, or infinite market depth). Realizing this, the insider would have incentive to demand in the first auction any quantity $\Delta \tilde{x}_1 \neq \Delta x_1$ in order to induce mispricing in the second auction and make unbounded profits (with probability one). Therefore, no invertible trading strategy can be part of an equilibrium with trade disclosure.

However, an equilibrium exists and at any auction the insider's trading strategy is determined by two components: (i) a private information-based linear component, $\beta_n \Delta t_n (v - p_{n-1}^*)$, and (ii) a noise component, normally and independently distributed, with zero mean and variance $\sigma_{z_n}^2 \Delta t_n$. Without loss of generality, let us consider in what follows the case $\Delta t_n = \Delta t \quad \forall n$.

THEOREM 1: *There exists a recursive linear equilibrium in which the constants β_n , λ_n , γ_n , α_n , δ_n , $\sigma_{z_n}^2$, and Σ_n characterize the following:*

$$\Delta x_n = \beta_n M_{n-1} \Delta t + \Delta z_n \quad (5)$$

$$M_{n-1} \equiv v - p_{n-1}^* \quad (6)$$

$$\Delta z_n \sim \mathcal{N}(0, \sigma_{z_n}^2 \Delta t) \quad (7)$$

$$p_n = p_{n-1}^* + \lambda_n \Delta y_n \quad (8)$$

$$p_n^* = p_{n-1}^* + \gamma_n \Delta x_n \quad (9)$$

$$\Sigma_n = \text{Var}[v | \mathcal{F}'_n] \quad (10)$$

$$J(W_n) = -\delta_{n-1} \exp\{-A (W_n + \alpha_{n-1} (v - p_{n-1}^*)^2)\} \quad (11)$$

$$W_n = W_{n-1} + \Delta x_{n-1} (v - p_{n-1}) \quad (12)$$

for all auctions $n \in \{1, \dots, N\}$. Given the initial condition $p_0^* = p_0$, and denoting with λ the

positive root of the following polynomial, that satisfy the SOC of the insider maximization problem at the last trading round,

$$\lambda^4 \left[A^2 (\sigma_u^2 \Delta t)^3 \right] + \lambda^3 \left[4A (\sigma_u^2 \Delta t)^2 \right] + \lambda^2 \left[4\sigma_u^2 \Delta t \right] - \lambda \left[A \sigma_u^2 \Delta t \frac{\Sigma_0}{N} \right] - \frac{\Sigma_0}{N} = 0 \quad (13)$$

the constants λ_n , γ_n , α_n , δ_n , β_n , $\sigma_{z_n}^2$, Σ_n are given by

$$\lambda_n = \lambda \quad (14)$$

$$\gamma_n = \gamma = 2\lambda + A\lambda^2 \sigma_u^2 \Delta t \quad (15)$$

$$\delta_n = 1 \quad (16)$$

$$\beta_n = \left(\frac{\Lambda \sigma_u^2}{\Sigma_0} \right) \frac{N}{N - n + 1} \quad (17)$$

$$\sigma_{z_n}^2 = \sigma_u^2 \left(\frac{\Lambda}{\gamma} \right) \frac{N - n}{N - n + 1} \quad (18)$$

$$\Sigma_n = \Sigma_0 \frac{N - n}{N} \quad (19)$$

for all auctions $n \in \{1, \dots, N\}$, and by

$$\alpha_n = (2\gamma)^{-1} \quad (20)$$

$$\alpha_N = 0 \quad (21)$$

for all auctions $n \in \{1, \dots, N - 1\}$, where the constant Λ is defined as follows:

$$\Lambda = \gamma(1 + A\lambda \sigma_u^2 \Delta t)^{-1} \quad (22)$$

Proof. See the Appendix. □

Note that if we set $A = 0$ we obtain the same equilibrium as in Huddart et al. (2001), in which case the positive root of the polynomial in Equation (13) is equal to $\sqrt{\Sigma_0/4\sigma_u^2}$.

3 A Continuous Auction Equilibrium

Consider now a “more realistic continuous-time setting” (Bloomfield and O’Hara, 1999). Let $dx(t)$ and $du(t)$ be the orders placed respectively by the insider trader and by liquidity traders at time t . We then assume that the latter follows an arithmetic Brownian motion,

$$du(t) = \sigma_u dB^u(t) \quad (23)$$

for some constant σ_u .⁷ The total order flow at time t is given by $dy(t) = dx(t) + du(t)$. The concept of pre- and post-disclosure in continuous-time may become misleading to visualize. Therefore, we prefer to give a slightly different interpretation to the sequence of events, in order not have to deal with an event that occurs between two instants. At each instant t the insider trader simultaneously reveals the trade she has made a dt before and places the new order. Hence, at each instant t the market maker observes $dx(t - dt)$ and $dy(t)$; so, he now computes first $p^*(t)$ and then sets the market price $p(t)$.⁸ The information structure is then represented by the following filtrations:

$$\begin{aligned}\mathbb{F}' &= \{\mathcal{F}'(t)\}_{t \geq 0} \quad \text{where} \quad \mathcal{F}'(t) = \sigma(dy(s), dx(s) : 0 \leq s < t) \\ \mathbb{F} &= \{\mathcal{F}(t)\}_{t \geq 0} \quad \text{where} \quad \mathcal{F}(t) = \sigma(dy(s) : 0 \leq s \leq t) \vee \sigma(dx(s) : 0 \leq s < t) \\ \mathbb{G} &= \{\mathcal{G}(t)\}_{t \geq 0} \quad \text{where} \quad \mathcal{G}(t) = \sigma(v) \vee \sigma(p(s), p^*(s) : 0 \leq s < t)\end{aligned}$$

3.1 Equilibrium

As in the discrete-time model, we restrict our attention to linear equilibria, in which both the trading strategy and the pricing rules are linear functions.⁹

DEFINITION 2: *A linear equilibrium of the continuous-time trading game is defined by the following conditions:*

1. **UTILITY MAXIMIZATION.** *Given the linear pricing rules $p(t) = P_t(\mathcal{F}(t))$ and $p^*(t) = P_t^*(\mathcal{F}'(t))$, the linear trading strategy $dx(t) = X_t(\mathcal{G}(t))$ maximizes*

$$\mathbb{E}[-\exp\{-AW(1)\}] \tag{24}$$

⁷Back (1992), Back and Pedersen (1998), and Baruch (2002) consider a model in which the instantaneous variance changes over time. Since this feature would not add new insights to our result, we prefer to consider the standard case of constant variance.

⁸Note that this simply means that the disclosure event at auction n in the discrete-time model now coincides with the trading event at auction $n + 1$.

⁹Since the aim of this paper is not to (technically) generalize the Kyle (1985) model, we consider a locally linear choice space (space in which linear pricing rules and linear trading strategies are defined) in which doubling strategies are ruled out. We refer to Back (1992) for further details.

2. MARKET EFFICIENCY. *Given the linear trading strategy $dx(t) = X_t(\mathcal{G}(t))$, the pricing rules are competitive:*

$$p(t) = \mathbb{E}[v|\mathcal{F}(t)] \quad (25)$$

$$p^*(t) = \mathbb{E}[v|\mathcal{F}'(t)] \quad (26)$$

The next theorem presents the closed-form solution for the linear equilibrium of the trading game in the continuous-time framework.

THEOREM 2: *There exists a recursive linear equilibrium in which the constants $\lambda(t)$, $\gamma(t)$, $\beta(t)$, $\alpha(t)$, $\sigma_{z(t)}$, and $\Sigma(t)$ characterize the following:*

$$dx(t) = \beta(t)M(t)dt + dz(t) \quad (27)$$

$$M(t) \equiv v - p^*(t) \quad (28)$$

$$dz(t) = \sigma_{z(t)}dB^z(t) \quad \text{with} \quad d\langle B^z, B^u \rangle_t = 0 \quad (29)$$

$$p(t) = p^*(t) + \lambda(t)dy(t) \quad (30)$$

$$dp^*(t) = \gamma(t)dx(t) \quad (31)$$

$$\Sigma(t) = \text{Var}[v|\mathcal{F}'(t)] \quad (32)$$

$$J(M, t) = -\exp\{-A[W(t) + V(M, t)]\} \quad (33)$$

$$V(M, t) = \alpha(t)[M(t)]^2 \quad (34)$$

$$W(t) = \int_0^t (v - p(s))dx(s) \quad (35)$$

for every $t \in [0, 1]$. Given the initial condition $p^*(0) = p(0)$, the constants $\beta(t)$, $\lambda(t)$, $\gamma(t)$, $\alpha(t)$, $\sigma_{z(t)}$, $\Sigma(t)$ are given by

$$\Sigma(t) = (1 - t)\Sigma(0) \quad (36)$$

$$\gamma(t) = \hat{\gamma} = (\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (37)$$

$$\lambda(t) = \frac{1}{2}\hat{\gamma} = \frac{1}{2}(\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (38)$$

$$\beta(t) = \hat{\gamma}^{-1}/(1 - t) = \sigma_u\Sigma(0)^{-\frac{1}{2}}/(1 - t) \quad (39)$$

$$\alpha(t) = \frac{1}{2}\hat{\gamma}^{-1} = \frac{1}{2}(\sigma_u^2/\Sigma(0))^{\frac{1}{2}} \quad (40)$$

$$\sigma_{z(t)}^2 = \sigma_u^2, \quad t \in (0, 1) \quad (41)$$

with the boundary condition $\sigma_{z(1)} = 0$.

Proof. See the Appendix. □

A somewhat surprising feature of this equilibrium is that it is independent of the level of risk aversion. This and other properties are discussed in detail in the next Section.

4 Properties of the equilibria

As in Kyle (1985), the parameter Σ_n , which gives the error variance of prices after the n -th auction, is an inverse measure of the informational efficiency of prices, which indicates how much of the insider's private information is not yet incorporated into prices. The parameter λ_n , which characterizes the market maker pricing function, is an inverse measure of market liquidity (or market depth). The parameter β_n , which characterizes the insider's trading strategy, represents the intensity of insider trading (or the informed trader aggressiveness), that is the sensitivity of the informed order flow to private information. Clearly, the continuous-time counterparts of these variables maintain the same meaning. In what follows we first discuss the equilibrium in discrete-time and subsequently the one in continuous-time.

Sequential auction equilibrium. To illustrate the effect of mandatory insider trade disclosure in a risk averse world, we present some numerical simulations in which each of the above mentioned market component is plotted over time. In all the simulations, we normalize exogenous parameters by setting $\Sigma_0 = 1$, $\sigma_u^2 = 1$, and $\Delta t = 1/N$.

To better highlight our results Panel A of Table 1 shows the effect of the disclosure regulation when the insider trader is assumed risk neutral. The three graphs contrast respectively Σ_n , λ_n , and β_n , (i) in a transparent market (solid line), and (ii) in an opaque market (dashed line). This means plotting Huddart et al. (2001) versus Kyle (1985). The three graphs clearly show that when the insider must disclose her trade at the end of each auction (we consider $N = 20$) the error variance of price and market depth are smaller, while the insider trading intensity is higher if compared to the case in which no such regulation is imposed. Indeed, these three graphs summarize the main result of Huddart et al. (2001),

according to which information is reflected more rapidly in prices with disclosure of insider trades than without. For this reason they support the enforcement of such regulation.

If we now move to our risk averse world, Panel B of Table 1 presents the same graphs as in Panel A but considering an informed agent characterized by a CARA utility function with coefficient of risk aversion $A = 4$. It is striking that the above results are reversed. Σ_n declines very rapidly through time in an opaque market and it is lower than the conditional variance in a transparent market. As far as the intertemporal pattern of market depth is concerned, the second graph shows that λ_n is constant with disclosure, while it declines quite sharply over time, becoming even lower in the last trading periods, without disclosure. Finally, in a transparent market insider trading intensity is lower: a lower aggressiveness reduces the efficiency of the market. Let us consider the case of 20 auctions and coefficient of risk aversion $A = 4$. In a transparent market, about 60% of the private information is incorporated into price just after a quarter of the total trading period. In an opaque market, only 25%. This strong difference even widens if the insider is more risk averse or if the number of auctions increases.¹⁰

Continuous auction equilibrium. The intuitions behind the equilibrium with continuous auctions are the same as those just discussed for the equilibrium in the discrete-time framework. However, the closed-form solution of the continuous-time model allows us to formally characterize our results. Before stating the main propositions, let us briefly comment the graphs in Table 2 and 3.

In Table 2 each graph is the exact counterpart of those presented in Table 1 for the limiting case $\Delta t \rightarrow 0$. The main difference lies in the first graph in Panel A: if trading is continuous, even when traders are risk neutral a disclosure regulation is not able to improve

¹⁰If $A = 10$, about 70% of the private information is incorporated into price in a transparent market, against 25% in an opaque market. If $N = 100$ more than 80% against 25%. If we combine both, about 95% still against 25% in the opaque market. Further simulations and comparative dynamics are available upon request.

market efficiency. When, instead, agents are risk averse, the enforcement of ex-post insider trade disclosure causes the market to be highly inefficient. In a transparent market, the instantaneous change in prior variance, $d\Sigma(t)$, is constant for any level of risk aversion. This implies that trading price has constant volatility over time and therefore that information is incorporated into prices at a constant rate. Surprisingly this is exactly the same path we would observe in an opaque market with a risk neutral informed agent (Kyle, 1985, Equations (4.9) and (4.10) pg. 1328). As a matter of fact, an interesting feature of our model is that the equilibrium is independent of the level of the risk aversion. The reader may convince himself just by observing the equilibrium equations in Theorem 2. Therefore, such regulation indirectly forces the insider to behave as a risk neutral monopolist, which we know that represents the most harmful scenario in terms of price efficiency. In an opaque market, instead, a risk averse informed agent would trade very aggressively at the early stages of the trading period in order to protect herself against future price risk: the leakage of private information through market prices improves informational efficiency, as shown by the first graph in Panel B. The same result can be seen from the perspective of the informed agent: insider trading behavior is less aggressive in an opaque market, as show the third graph in Panel B. Let $\tilde{\omega}(t)$ denote the equilibrium path of a generic variable ω in an opaque market. Then, the following proposition summarizes our result on market efficiency.

PROPOSITION 1: *For any level of risk aversion, the insider trader is more aggressive, $\tilde{\beta}(t) \geq \beta(t)$, and the market is more efficient at any point in time, $\tilde{\Sigma}(t) \leq \Sigma(t)$, in an opaque market.*

Proof. See the Appendix. □

Another important result is that also market liquidity may decrease in an transparent market. In particular, in order to sustain the insider's optimal mixed strategy, in equilibrium market liquidity must be constant over time. If this were not the case, the insider would have an incentive to deviate from that strategy and exploit a lower trading cost. At the

early stages of the trading period the adverse selection problem faced by the market maker is reduced, implying a higher market depth. However, at the late stages of the trading period it may be the case that liquidity is higher in an opaque market, as shown by the second graph in Panel B. The following proposition summarizes our result on market liquidity.

PROPOSITION 2: *If the informed agent is sufficiently risk averse ($A > A^*$), then market liquidity at the end of the trading period is higher in an opaque market. The higher the risk aversion with respect to A^* , the longer the trading sub-period in which market liquidity is lower in a transparent market.*

Proof. See the Appendix. □

Finally, in Table 3 we show how the results described in Table 2 depend on the risk attitude of the informed agent by adding a third dimension to the previous graphs. Now the smooth surface represents the transparent market, whereas the striped surface the opaque one. As already mentioned, the higher the level of risk aversion, the stronger the detrimental effect of a higher degree of transparency on market efficiency and liquidity. The three graphs make this point very clear.

5 Conclusions and Policy Implications

Securities markets worldwide have different degrees of transparency with implications that are not well understood. In this article we focus on mandatory ex-post trade disclosure by corporate insiders as a particular aspect of market transparency. In a model of risk averse strategic insider trading we show that in a transparent market (i.e. with disclosure regulation) informational efficiency and market liquidity are significantly lower than in an opaque market (i.e. without disclosure regulation). The reason for this detrimental effect is that a risk averse insider optimally chooses a less aggressive trading strategy in order to prevent the market maker from inferring perfectly the private information from public records, and

to maintain her informational advantage over time. To easily guide the reader through the main feature of the model we have firstly presented a sequential auction equilibrium and then derived the main results in closed-form relying on a more elegant continuous-time setting.

Our result adds an interesting theoretical evidence to the existing debate on the relation between the degree of transparency and the optimal design and regulation of securities markets. A number of implications for regulatory policy can be drawn from our analysis. First, a mandatory insider trade disclosure does not eliminate the presence of insider trading once the informed agent reveals her trade: this constitutes a relevant empirical implication to test. Second, if the main goal of market design is to sustain informational efficiency and liquidity, as clearly stated for example in Section 16(a) of the U.S. Securities Exchange Act, then an opaque market should be preferred. Strategic corporate insiders would exploit their informational advantage more rapidly, inducing private information to reach the market (and hence to be available to all investors) on a more timely basis. Under this view a trade disclosure regulation would represent a friction in the system of efficient prices. Finally, such regulation may not be able to sufficiently reduce the adverse selection problem faced by the market maker, and consequently to enhance market liquidity. Testable implications on insiders' trading behavior and on market components then follow.

The change in the U.S. regulation on insider trade disclosure with the introduction of the Sarbanes-Oxley Act in 2002 offers the opportunity to carry out a natural experiment to test the implication derived in this article. Moreover, having shown how the existing regulation fails at achieving its goals, then the next natural step is to study and endogenously derive what should be the optimal regulation. These interesting and challenging projects are left for further research.

Appendix

PROOF OF THEOREM 1: We proceed by backward induction. Suppose that for constants α_n and δ_n

$$J(W_{n+1}) = -\delta_n \exp[-A(W_{n+1} + \alpha_n M_n^2)] \quad (\text{A1})$$

We have then

$$\begin{aligned} J(W_n) &= \max_{\Delta x} \mathbb{E}[J(W_{n+1})|\mathcal{G}_n] \\ &= \max_{\Delta x} \mathbb{E}_n[-\delta_n \exp[-A(W_n + \Delta x(v - p_n) + \alpha_n M_n^2)]] \end{aligned} \quad (\text{A2})$$

$$\text{s.t.} \quad \begin{cases} p_n &= p_{n-1}^* + \lambda_n(\Delta x + \Delta u_n) \\ p_n^* &= p_{n-1}^* + \gamma_n \Delta x \end{cases} \quad (\text{A3})$$

where Δx denotes the control quantity of the insider trader at the n -th auction, and the two constraints highlight the linear functional form of the conjectured pricing functions. Substituting Equations (A3) in (A2), which clearly shows the strategic behavior of the insider, and evaluating the conditional expectation for log-normal distributions, we obtain

$$J(W_n) = \max_{\Delta x} -\delta_n \exp \left\{ -A \left[W_n + \Delta x [1 - 2\alpha_n \gamma_n] M_{n-1} + (\Delta x)^2 \left[\alpha_n \gamma_n^2 - \lambda_n - \frac{1}{2} A \lambda_n^2 \sigma_u^2 \Delta t \right] + \alpha_n M_{n-1} \right] \right\} \quad (\text{A4})$$

The first-order condition (FOC), where Δx_n denote the optimized value of Δx for the above expression, turns out to be

$$M_{n-1} [1 - 2\alpha_n \gamma_n] + \Delta x_n [2\alpha_n \gamma_n^2 - 2\lambda_n - A \lambda_n^2 \sigma_u^2 \Delta t] = 0 \quad (\text{A5})$$

Since the conjectured trading strategy of the informed agent incorporates a random component (the dissimulation part of the trade), it means that in equilibrium the insider must be indifferent across all possible values of her trade, that is the FOC must hold for any Δx_n . This implies that the following conditions (hereafter called *dissimulation conditions*) must hold for any $n \in \{1, \dots, N-1\}$:

$$1 - 2\alpha_n \gamma_n = 0 \quad (\text{A6})$$

$$\alpha_n \gamma_n^2 - \lambda_n - \frac{1}{2} A \lambda_n^2 \sigma_u^2 \Delta t = 0 \quad (\text{A7})$$

Note that, if these conditions are satisfied, both FOC and SOC hold.

In equilibrium the conjectured indirect utility must be correct. This means that α_{n-1} and δ_{n-1} must be such that Equation (A1) at time t_n is equal to Equation (A2):

$$-\delta_{n-1} \exp[-A(W_n + \alpha_{n-1} M_{n-1}^2)] = \max_{\Delta x} \mathbb{E}[J(W_{n+1})|\mathcal{G}_n] \quad (\text{A8})$$

When computing the expectation in the r.h.s. of the above expression we must take into account that dissimulation of insider trading implies randomness in the insider trading strategy. In particular, $J(W_{n+1})$ contains linear and quadratic functions of two independent (normally distributed) random components: Δu_n and Δz_n . For this purpose let us consider the following lemma.

LEMMA 1: *Let X and Z be two independent normally distributed random variables: $X \sim \mathcal{N}(0, \sigma_X^2)$, $Z \sim \mathcal{N}(0, \sigma_Z^2)$. Then, if $\sigma_X^2(2a + c^2\sigma_Z^2) < 1$:*

$$\mathbb{E}[\exp(aX^2 + bX + dZ + cXZ)] = \exp\left(\frac{\Psi}{2\Phi}\right) \frac{1}{\sqrt{\Phi}} \quad (\text{A9})$$

where

$$\Psi \equiv d^2\sigma_Z^2 + \sigma_X^2(b^2 + 2d\sigma_Z^2(bc - ad)) \quad (\text{A10})$$

$$\Phi \equiv 1 - \sigma_X^2(2a + c^2\sigma_Z^2) \quad (\text{A11})$$

If $\sigma_X^2(2a + c^2\sigma_Z^2) > 1$, the above expectation is not well-defined.

Proof. See Brunnermeier (2001, pag. 64). □

Simple algebra leads to the following expression for the indirect utility at t_n :

$$J(W_n) = -\delta_n \left[\exp(-AQ_n) \mathbb{E}_n \left[\exp\left(a_n(\Delta z_n)^2 + b_n\Delta z_n + d_n\Delta u_n + c_n\Delta z_n\Delta u_n\right) \right] \right] \quad (\text{A12})$$

where

$$\begin{cases} Q_n & \equiv W_n + [\beta_n(1 - 2\alpha_n\gamma_n) + \beta_n^2(\alpha_n\gamma_n^2 - \lambda_n) + \alpha_n]M_{n-1}^2 \\ a_n & \equiv -A(\alpha_n\gamma_n^2 - \lambda_n) \\ b_n & \equiv -A[(1 - 2\alpha_n\gamma_n) + 2\beta_n(\alpha_n\gamma_n^2 - \lambda_n)]M_{n-1} \\ d_n & \equiv A\lambda_n\beta_nM_{n-1} \\ c_n & \equiv A\lambda_n \end{cases} \quad (\text{A13})$$

Using Lemma 1 and the two *dissimulation conditions* in Equations (A6) and (A7), we find that α and δ are constant over time until the second last auction. Therefore, since

$$\alpha_n = [2\lambda_n(2 + A\lambda_n\sigma_u^2\Delta t)]^{-1} \quad \text{and} \quad \alpha_{n-1} = \alpha_n \quad \forall n \in \{1, \dots, N-1\} \quad (\text{A14})$$

it follows that

$$\gamma_{n-1} = \gamma_n \quad \forall n \in \{1, \dots, N-1\} \quad (\text{A15})$$

$$2(\lambda_n - \lambda_{n-1}) + A\sigma_u^2\Delta t(\lambda_n^2 - \lambda_{n-1}^2) = 0 \quad (\text{A16})$$

which implies

$$\lambda_{n-1} = \lambda_n \quad \forall n \in \{1, \dots, N-1\} \quad (\text{A17})$$

Solving the maximization problem that the insider faces at the last trading round, in which it is clearly optimal not to add any noise component to the order flow, we get as in Holden and Subrahmanyam (1994) the following equations:

$$\beta_N \Delta t = [\lambda_N(2 + A\lambda_N\sigma_u^2\Delta t)]^{-1} \quad (\text{A18})$$

$$\alpha_{N-1} = [2\lambda_N(2 + A\lambda_N\sigma_u^2\Delta t)]^{-1} \quad (\text{A19})$$

$$\delta_{N-1} = 1 \quad (\text{A20})$$

Finally, Equation (14) follows from Equation (A14) evaluated at second last auction and from Equation (A19). As in Huddart et al. (2001) market depth is constant over all trading rounds. Moreover, since no utility can be gained after trading is complete, $\alpha_N = 0$ and $\delta_N = 1$.

Market efficiency conditions requires that $p_n = \mathbb{E}[v|\mathcal{F}_n]$ and $p_n^* = \mathbb{E}[v|\mathcal{F}'_n]$. Simple application of the projection theorem for normally distributed random variables confirms the linear pricing rules specified in Equations (8) and (9), where the slope coefficients are respectively given

$$\lambda_n = \frac{\beta_n \Sigma_{n-1}}{\beta_n^2 \Delta t \Sigma_{n-1} + \sigma_{z_n}^2 + \sigma_u^2} \quad (\text{A21})$$

$$\gamma_n = \frac{\beta_n \Sigma_{n-1}}{\beta_n^2 \Delta t \Sigma_{n-1} + \sigma_{z_n}^2} \quad (\text{A22})$$

By the same theorem and using Equation (A22) we find the expression for the conditional variance:

$$\Sigma_n = \Sigma_{n-1}(1 - \gamma_n \beta_n \Delta t) \quad (\text{A23})$$

Furthermore, simple algebra allows us to conclude that also γ_n is constant for any auction $n \in \{1, \dots, N\}$.

Now, combining Equations (A21) and (A22) we can derive a convenient expression for β_n :

$$\beta_n = \Lambda \sigma_u^2 (\Sigma_{n-1})^{-1} \quad (\text{A24})$$

where Λ is a constant defined in Equation (22). Substituting β_n in Equation (A23) we clearly see that the conditional variance decreases at a constant rate:

$$\Sigma_n = \Sigma_{n-1} - (\gamma \Lambda \sigma_u^2 \Delta t) \quad (\text{A25})$$

$$= \Sigma_0 - (\gamma \Lambda \sigma_u^2 \Delta t) n \quad \forall n \in \{1, \dots, N\} \quad (\text{A26})$$

Since all the private information is incorporated at the end the trading game, $\Sigma_N = 0$. Therefore,

$$0 = \Sigma_0 - (\gamma \Lambda \sigma_u^2 \Delta t) N \quad (\text{A27})$$

$$\Sigma_n = \Sigma_0 (N - n) / N \quad \forall n \in \{1, \dots, N\} \quad (\text{A28})$$

Equations (17) and (18) are obtained through straightforward algebra.

Finally, in order to obtain the optimal constant λ we need to solve the polynomial of degree four in Equation (13), obtained by combining Equations (A18), (A19) and (A28). Note that any positive root satisfies the SOC of the insider maximization problem at the last trading round: $-2\lambda - A\lambda^2\sigma_u^2\Delta t \leq 0$. This complete the proof of the theorem. $\square$

PROOF OF THEOREM 2: The objective of the insider trader is to maximize

$$\max_{\{dx(t)\}_t} \mathbb{E}[-\exp\{-AW(1)\}] = \max_{\{dx(t)\}_t} \mathbb{E}\left[-\exp\left\{-A \int_0^1 (v - p(s))dx(s)\right\}\right] \quad (\text{A29})$$

given the dynamic of the conjectured pricing rule $p(t)$ defined in Equations (30). This is a Markovian stochastic control problem:

$$J(M, t) = \max_{dx} \mathbb{E}_t[J(M, t + dt)] \quad (\text{A30})$$

$\Updownarrow$

$$-\exp\{-A[W(t) + V(M, t)]\} = \max_{dx} \mathbb{E}_t[-\exp\{-A[W(t + dt) + V(M, t + dt)]\}] \quad (\text{A31})$$

Exploiting the exponential function form, we get a very convenient expression just by dividing both sides of the above equation by the LHS:

$$1 = \max_{dx} \mathbb{E}_t[\exp\{-A[dW(t) + dV(M, t)]\}] \quad (\text{A32})$$

Application of Ito's Lemma allows us to identify the following dynamics:¹¹

$$dW(t) = (v - p(t))dx(t) \quad (\text{A33})$$

$$= M(t)dx(t) - \lambda(t)d\langle x, x \rangle_t - \lambda(t)d\langle u, x \rangle_t \quad (\text{A34})$$

$$= M(t)dx(t) - \lambda(t)d\langle x, x \rangle_t \quad (\text{A35})$$

$$dM(t) = -\gamma(t)dx(t) \quad (\text{A36})$$

$$d\langle M, M \rangle_t = \gamma(t)^2 d\langle x, x \rangle_t \quad (\text{A37})$$

$$dV(M, t) = V_M dM(t) + \frac{1}{2}V_{MM}d\langle M, M \rangle_t + V_t dt \quad (\text{A38})$$

$$= [2\alpha(t)M(t)]dM(t) + [\alpha(t)]d\langle M, M \rangle_t + M^2(t)d\alpha(t) \quad (\text{A39})$$

Given that dx is a control variable, $[dW(t) + dV(M, t)]$ is not random. Therefore, we can rewrite Equation (A32) as

$$1 = \max_{dx} \exp\left\{-A\left[dx\left[1 - 2\alpha(t)\gamma(t)\right]M(t) + (dx)^2\left[\alpha(t)\gamma(t)^2 - \lambda(t)\right] + M^2(t)d\alpha(t)\right]\right\} \quad (\text{A40})$$

¹¹The differential $d\langle H, K \rangle_t$ simply coincide with $dH(t)dK(t)$ and represents the differential of the quadratic variation process. Note that $d\langle u, x \rangle_t = 0$ since $d\langle B^u, B^z \rangle_t = 0$ (the two Brownian motion are assumed independent) and $dB^u dt$ is of order greater than dt .

The first-order condition (FOC) then is

$$\left[1 - 2\alpha(t)\gamma(t)\right]M(t) + 2dx(t)\left[\alpha(t)\gamma(t)^2 - \lambda(t)\right] = 0 \quad (\text{A41})$$

Exactly as in the discrete-time framework, because the conjectured trading strategy of the informed agent incorporates a random component, in equilibrium the insider must be indifferent across all possible values of her trade, that is the FOC must hold for any $dx(t)$. The following *dissimulation conditions* must hold for any $t \in (0, 1)$:

$$1 - 2\alpha(t)\gamma(t) = 0 \quad (\text{A42})$$

$$\alpha(t)\gamma(t)^2 - \lambda(t) = 0 \quad (\text{A43})$$

Note that the FOC and the dissimulation conditions we obtain are independent of the level of risk aversion. Evaluating Equation (A40) with the above optimal conditions we get

$$1 = \exp\left\{-A\left[M^2(t)d\alpha(t)\right]\right\} \quad (\text{A44})$$

which holds for every level of $M^2(t)$ and A if and only if $\alpha(t)$ is constant:

$$d\alpha(t) = 0 \Rightarrow \alpha(t) = \hat{\alpha} \quad \forall t \quad (\text{A45})$$

From Equation (A42) if $\alpha(t)$ is constant, it must be the case that also $\gamma(t)$ is constant. Moreover, from Equation (A43) if $\alpha(t)$ and $\gamma(t)$ are constant, it must be the case that $\lambda(t)$ is constant as well. Therefore,

$$\hat{\lambda} = \frac{1}{2}\hat{\gamma} \quad (\text{A46})$$

$$\hat{\alpha} = \frac{1}{2}\hat{\gamma}^{-1} \quad (\text{A47})$$

Following closely Back et al. (2000) we now define in the following Lemma the filtering problem of the market maker.

LEMMA 2: Assume the insider trader follow the linear strategy as in Equation (27) and let us define $V(t) = \mathbb{E}[v|\mathcal{F}'(t)]$ and $\Sigma(t) = \mathbb{V}\text{ar}[v|\mathcal{F}'(t)]$. Then the following process

$$Q(t) = \int_0^t \beta(s)[v - V(s)] + z(s) \, ds \quad (\text{A48})$$

is a Wiener process on the market maker's information structure $\mathbb{F}'$ and it is called “innovation” process for the market maker's estimation problem. The differential

$$dQ(t) = \beta(t)[v - V(t)]dt + dz(t) \quad (\text{A49})$$

is the unpredictable part of the order flow from the informed agent. Moreover,

$$V(t) = \int_0^t \frac{\beta(s)\Sigma(s)}{\sigma_{z(s)}^2} dQ(s) \quad (\text{A50})$$

Therefore, the market maker's estimate of v is revised according to

$$dV(t) = \frac{\beta(t)\Sigma(t)}{\sigma_{z(t)}^2} dQ(t) \quad (\text{A51})$$

Finally,

$$d\Sigma(t) = -\frac{[\beta(t)\Sigma(t)]^2}{\sigma_{z(t)}^2} dt \quad (\text{A52})$$

Proof. This is an application of the Kalman-Bucy filter. See Kallianpur (1980, Sec. 10.3). $\square$

Comparing Equation (A51) with Equation (31) we easily see that

$$\gamma(t) = \frac{\beta(t)\Sigma(t)}{\sigma_{z(t)}^2} \quad (\text{A53})$$

Applying the same logic as in Lemma 2, but considering the market maker's information structure $\mathbb{F}$, we also obtain that

$$\lambda(t) = \frac{\beta(t)\Sigma(t)}{\sigma_{z(t)}^2 + \sigma_u^2} \quad (\text{A54})$$

The optimal dissimulation component is found by considering that $\hat{\lambda} = \frac{1}{2}\hat{\gamma}$:

$$\frac{\beta(t)\Sigma(t)}{\sigma_{z(t)}^2 + \sigma_u^2} = \frac{1}{2} \frac{\beta(t)\Sigma(t)}{\sigma_{z(t)}^2} \quad (\text{A55})$$

$\Downarrow$

$$\sigma_{z(t)}^2 = \sigma_u^2 \quad (\text{A56})$$

Given the constant drift for the conditional variance process,

$$\Sigma(t) = \Sigma(0) - \int_0^t \gamma(s)^2 \sigma_{z(s)}^2 ds \quad (\text{A57})$$

$$= \Sigma(0) - (\hat{\gamma}^2 \sigma_u^2) t \quad \forall t \in [0, 1] \quad (\text{A58})$$

and, given our linear choice space, $\lim_{s \rightarrow 1} p^*(s) = \lim_{s \rightarrow 1} p(s) = v$, all the private information is incorporated into prices at the end of the trading game: $\Sigma(1) = 0$. Then, it follows that

$$\hat{\gamma} = (\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (\text{A59})$$

All the other constants $(\beta(t), \lambda(t), \alpha(t))$ are obtained through straightforward algebra. The boundary conditions $\alpha(1) = 0$ and $\sigma_{z(1)} = 0$ states respectively that no utility can be gained after trading is complete and that no dissimulation component is added to the trading strategy at the last instant. This complete the proof of the theorem. $\square$

PROOF OF PROPOSITION 1: Following Baruch (2002), the conditional variance of prices and the insider trading intensity in a continuous-time model with risk averse informed trader in an opaque market (no disclosure regulation) are defined by

$$\tilde{\Sigma}(t) = \frac{-\Sigma(0) \left[2 + A^2 \Sigma(0) \sigma_u^2 - \sqrt{A^2 \Sigma(0) \sigma_u^2 (4 + A^2 \Sigma(0) \sigma_u^2)} \right] (1-t)t}{-[2 + 2A^2 \Sigma(0) \sigma_u^2 (1-t)t]} \quad (\text{A60})$$

$$\tilde{\beta}(t) = \frac{2\sigma_u}{\left[\sqrt{\Sigma(0)(4 + A^2 \Sigma(0) \sigma_u^2)} - A \Sigma(0) \sigma_u \right] (1-t)} \quad (\text{A61})$$

Then, $\Sigma(t) - \tilde{\Sigma}(t) \geq 0$ and $\tilde{\beta}(t) - \beta(t) \geq 0$ imply respectively

$$\frac{-A[\Sigma(0)]^{\frac{3}{2}} \sigma_u \left[\sqrt{4 + A^2 \Sigma(0) \sigma_u^2} + \sqrt{A^2 \Sigma(0) \sigma_u^2 (1-2t)} \right] (1-t)t}{-[2 + 2A^2 \Sigma(0) \sigma_u^2 (1-t)t]} \geq 0 \quad (\text{A62})$$

$$\frac{A \Sigma(0) \sigma_u^2 + \sqrt{\Sigma(0)} \sigma_u \left[\sqrt{4 + A^2 \Sigma(0) \sigma_u^2} - 2 \right]}{2 \Sigma(0) (1-t)} \geq 0 \quad (\text{A63})$$

Straightforward algebra can show that these two inequalities hold true for any $t \in [0, 1]$. $\square$

PROOF OF PROPOSITION 2: Following Baruch (2002), market liquidity in a continuous-time model with risk averse informed trader in an opaque market is defined by

$$\tilde{\lambda}(t) = \left[\frac{\sigma_u \sqrt{4 + A^2 \Sigma(0) \sigma_u^2}}{2 \sqrt{\Sigma(0)}} + A \sigma_u^2 \left(t - \frac{1}{2} \right) \right]^{-1} \quad (\text{A64})$$

Since $\tilde{\lambda}(t)$ is decreasing over time,

$$\frac{\partial \tilde{\lambda}(t)}{\partial t} = - \frac{4A \Sigma(0)}{\left(\sqrt{4 + A^2 \Sigma(0) \sigma_u^2} + A \sigma_u \sqrt{\Sigma(0)} (2t - 1) \right)^2} \quad (\text{A65})$$

this means that $\tilde{\lambda}(t)$ and $\lambda(t)$ cross at most once in $t \in [0, 1]$. Moreover, since $\tilde{\lambda}(1)$ is decreasing in the level of risk aversion,

$$\left. \frac{\partial \tilde{\lambda}(t)}{\partial A} \right|_{t=1} = - \frac{\Sigma(0)}{2} \left(1 - \sqrt{\frac{A^2 \Sigma(0) \sigma_u^2}{4 + A^2 \Sigma(0) \sigma_u^2}} \right) \quad (\text{A66})$$

then for any $A > A^*$, with $A^* = 3 \left[2 \sqrt{\Sigma(0) \sigma_u^2} \right]^{-1}$ market liquidity at the end of the trading period is higher in an opaque market. The second part of the proposition follows. $\square$

References

- [1] Back K., 1992, Insider Trading in Continuous Time, *Review of Financial Studies* 5(3), 387–409.
- [2] Back K., H.H. Cao, and G.A. Willard, 2000, Imperfect Competition Among Informed Traders, *Journal of Finance* 55, 2117–2155.
- [3] Back K., H. Pedersen, 1998, Long-lived information and intraday patterns, *Journal of Financial Market* 1, 385–402.
- [4] Baruch S., 2001, Insider Trading and Risk Aversion, *Journal of Financial Market* 5, 451–464.
- [5] Bloomfield R. and M. O’Hara, 1999, Market Transparency: Who Wins and Who Loses?, *Review of Financial Studies* 12, 5–35.
- [6] Brunnermeier M.K., 2001, *Asset Pricing under Asymmetric Information*, Oxford, UK: Oxford University Press.
- [7] Cao H.H. and Y. Ma, 1999, Trade Disclosure and Imperfect Competition among Insiders, working paper.
- [8] Fidrmuc J., M. Goergen and L. Renneboog, 2006, Insider Trading, News Releases and Ownership Concentration, *Journal of Finance* 61(6), 2931–2973.
- [9] Fishman M.J. and K.M. Hagerty, 1995, The Mandatory Disclosure of Trades and Market Liquidity, *Review of Financial Studies* 8, 637–676.
- [10] Flood M., R. Huisman, K. Koedijk and R. Mahieu, 1999, Quote Disclosure and Price Discovery in Multiple-Dealer Financial Markets, *Review of Financial Studies* 12, 37–59.
- [11] Foster F.D. and S. Viswanathan, 1996, Strategic Trading when Agents Forecast the Forecasts of Others, *Journal of Finance* 51, 1437–1478.
- [12] Frutos M.A. de and C. Manzano, 2002, Risk Aversion, Transparency, and Market Performance, *Journal of Finance* 57(2), 959–984.
- [13] Holden C.W. and A. Subrahmanyam, 1994, Risk Aversion, Imperfect Competition, and Long-lived Information, *Economics Letters* 44, 181–190.
- [14] Huddart S., J.S. Hughes and C.B. Levine, 2001, Public Disclosure and Dissimulation of Insider Traders, *Econometrica* 69, 665–681.
- [15] International Organization of Securities Commissions (IOSCO), 2001, *Transparency and Market Fragmentation*, Madrid.

- [16] John K. and R. Narayanan, 1997, Market Manipulation and the Role of Insider Trading Regulations, *Journal of Business* 70(2), 217–247.
- [17] Kallianpur G., 1980, *Stochastic Filtering Theory*, Springer-Verlag.
- [18] Kyle A.S., 1985, Continuous Auctions and Insider Trading, *Econometrica* 53, 1315–1335.
- [19] Madhavan A., 1995, Consolidation, Fragmentation, and the Disclosure of Trading Information, *Review of Financial Studies* 8, 579–603.
- [20] Naik N., A. Neuberger and S. Viswanathan, 1999, Trade Disclosure Regulation in Markets with Negotiated Trade, *Review of Financial Studies* 12, 886–900.
- [21] Securities and Exchange Commission (SEC), Division of Market Regulation, 1995, *Market 2000: An Examination of Current Equity Market Developments*, Washington, DC.
- [22] Securities and Investment Board (SIB), 1994, *Regulation of the United Kingdom Equity Markets*, London.
- [23] Zhang W.D., 2004, Risk Aversion, Public Disclosure, and Long-lived Information, *Economics Letters* 85, 327–334.

TABLE 1. SEQUENTIAL AUCTION EQUILIBRIA

The three graphs in each Panel contrast over time the conditional variance of prices (an inverse measure of market efficiency), Σ_n , the sensitivity of the pricing function to the total order flow (an inverse measure of market liquidity), λ_n , and the sensitivity of the insider's trading strategy to private information (a direct measure of insider aggressiveness) β_n , respectively for the discrete-time equilibrium (i) in a transparent market (solid line), and (ii) in an opaque market (dashed line). Panel A represents the case of risk neutrality ($A = 0$); Panel B the case of risk aversion ($A = 4$). Exogenous parameters are normalized by setting $\Sigma_0 = 1$, $\sigma_u^2 = 1$, and $\Delta t = 1/N$ with $N = 20$.

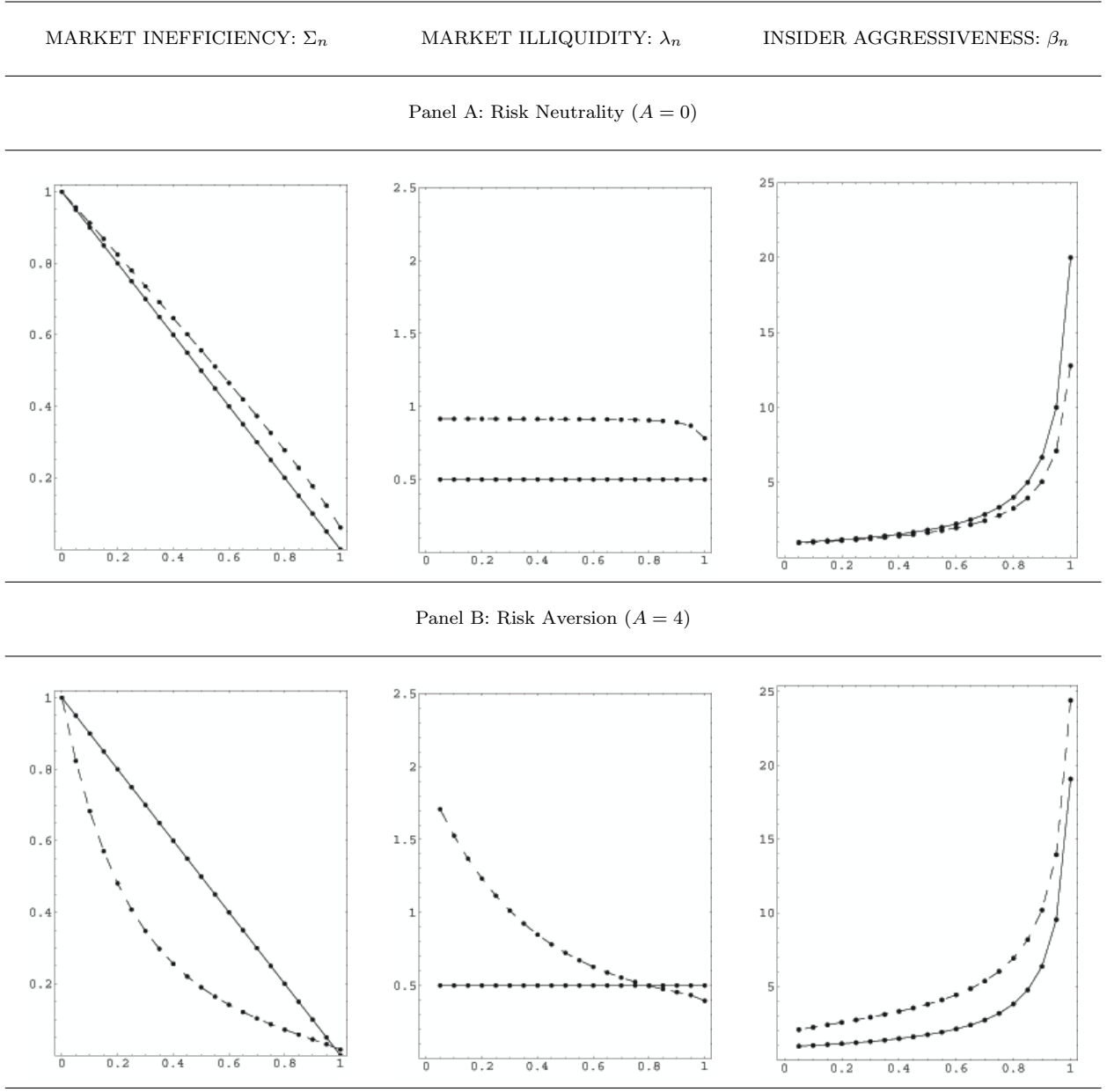


TABLE 2. CONTINUOUS AUCTION EQUILIBRIA I

The three graphs in each Panel contrast over time the conditional variance of prices (an inverse measure of market efficiency), $\Sigma(t)$, the sensitivity of the pricing function to the total order flow (an inverse measure of market liquidity), $\lambda(t)$, and the sensitivity of the insider's trading strategy to private information (a direct measure of insider aggressiveness) $\beta(t)$, respectively for the continuous-time equilibrium (i) in a transparent market (solid line), and (ii) in an opaque market (dashed line). Panel A represents the case of risk neutrality ($A = 0$); Panel B the case of risk aversion ($A = 4$). Exogenous parameters are normalized by setting $\Sigma(0) = 1$, $\sigma_u^2 = 1$.

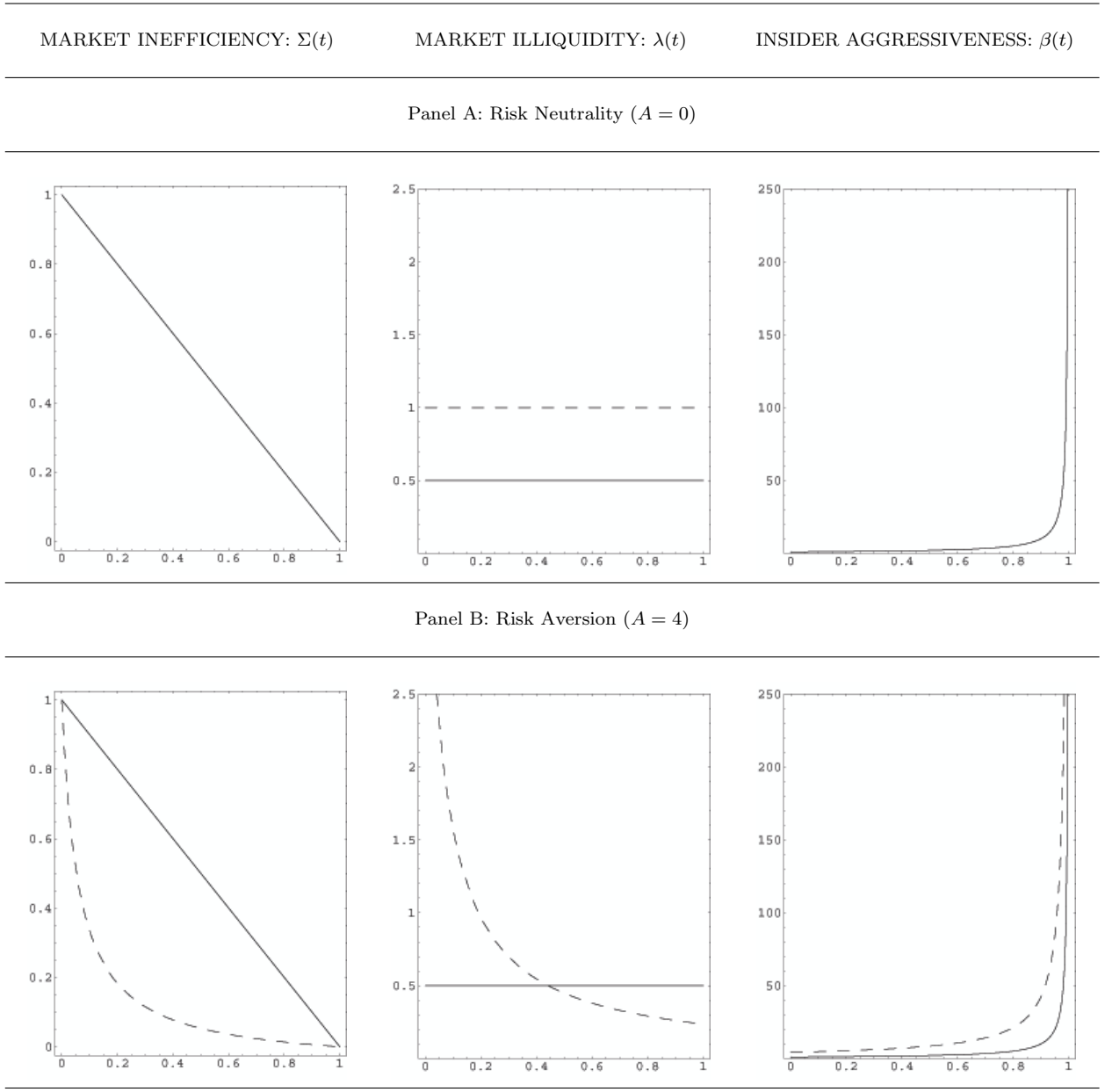


TABLE 3. CONTINUOUS AUCTION EQUILIBRIA II

The three graphs contrast over time and for different levels of risk aversion the conditional variance of prices (an inverse measure of market efficiency), $\Sigma(t, A)$, the sensitivity of the pricing function to the total order flow (an inverse measure of market liquidity), $\lambda(t, A)$, and the sensitivity of the insider's trading strategy to private information (a direct measure of insider aggressiveness) $\beta(t, A)$, respectively for the continuous-time equilibrium (i) in a transparent market (smooth surface), and (ii) in an opaque market (striped surface). Exogenous parameters are normalized by setting $\Sigma(0) = 1$, $\sigma_u^2 = 1$.

